Statistical hypothesis testing

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Outline

Introduction to tests

Tests on a single population

Tests on two populations

Notations

- $\mathcal{P}$ population
- X random variable
- $\theta \in \Theta$ unknown parameter ($\Theta =$ set of possible values for θ)
- if X is **continuous**: $x \mapsto f_\theta(x)$ density of X
- if X is **discrete**: $x \mapsto P_\theta(x)$ probability function of X i.e. $P_\theta(x) = \mathbb{P}(X = x)$
- $(X_1, \ldots, X_n)$ random (i.i.d.) sample of size n associated with X
- $(x_1, \ldots, x_n)$ observations in the sample

Objective

General scope

Take a decision regarding a question such as

- Is this new drug[1] more efficient?
- Is the life expectancy different between two groups of patients?
- Is the distribution of this random variable Gaussian? $\rightarrow$ yes/no question!

Statistical tests

$\rightarrow$ methodology to answer this kind of questions based on observations related to the phenomenon of interest

[1] "*médicament*"

Statistical tests: basic principles

- **Confronting two exclusive hypothesis**
  - the null hypothesis $\mathcal{H}_0$
  - the alternative hypothesis $\mathcal{H}_1$
- **Important:** the role of $\mathcal{H}_0$ and $\mathcal{H}_1$ are not symmetrical and totally different

  $\rightarrow$ $\mathcal{H}_0$ is assumed to be true to construct the test
- **The test decides either**
  - to keep $\mathcal{H}_0$: the result is then **not significant**[1]
  - to reject $\mathcal{H}_0$ in favor of $\mathcal{H}_1$: the result is then **significant**[2]

[1] "*non significatif*"
[2] "*significatif*"

Statistical tests: basic principles

Important

Answer	Meaning[3]
"keep $\mathcal{H}_0$"	~~yes~~ **maybe**
"reject $\mathcal{H}_0$"	no

- You can either **reject $\mathcal{H}_0$**[4] (i.e. validate that $\mathcal{H}_0$ is false) or say that the test is **not significant**.
- **A test can not confirm that $\mathcal{H}_0$ is true!**

[3] "*signification*"

[3] reject $\mathcal{H}_0$ = accept $\mathcal{H}_1$

Different categories of tests

- **Parametric test:** test regarding a parameter of a probabilility distribution under certain conditions
  - test on expectation X of mean μ, test $\mathcal{H}_0 : \mu = 5$ vs $\mathcal{H}_1 : \mu \neq 5$
  - test on variance X of variance σ^2, test $\mathcal{H}_0 : \sigma = 1$ vs $\mathcal{H}_1 : \sigma < 1$
  - test on proportion $X \sim \mathcal{B}(p)$, test $\mathcal{H}_0 : p = 0.5$ vs $\mathcal{H}_1 : p > 0.5$
- **Non-parametric test**
  - regarding generic parameters (mediane, quantile) independently from a probability distribution
  - regarding the probability distribution of a random variable (not just its parameter): test of adjustement to a distribution[1]
  - regarding the independance between two random variables: independence test

[1] "*test d'ajustement à une loi*"

General approach

1. Formulate the **hypotheses** $\mathcal{H}_0$ and $\mathcal{H}_1$

2. Find a **test statistic**[1] noted T_n, i.e. a quantity computed from the sample $(X_1, \ldots, X_n)$, whose probability distribution is explicit when assuming $\mathcal{H}_0$ true

3. Compute the **realization** t_n of T_n with the sample observations $(x_1, \ldots, x_n)$

4. **Reject** $\mathcal{H}_0$ if the **value** t_n **is improbable** regarding the distribution of T_n under $\mathcal{H}_0$

Note: The improbability of the value t_n is computed depending on a certain level α

[1] "*Statistique de test*"

Example

Context

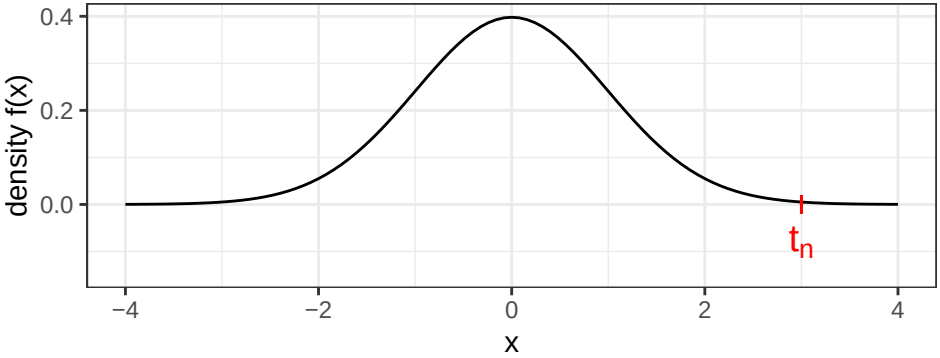
- X of mean μ and variance σ^2 measures a quantity on the population $\mathcal{P}$
- Question regarding the theoretical value of μ, e.g. "is μ equal to a given value μ_0?"
- n large, $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$

Procedure

1. Formulation of the test "$\mathcal{H}_0$: $\mu = \mu_0$" vs "$\mathcal{H}_1$: $\mu \neq \mu_0$"
2. test statistic $T_n = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$ under $\mathcal{H}_0$
3. Compute the observed $t_n = \frac{\bar{x}_n - \mu_0}{s/\sqrt{n}}$ where $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2$
4. Is the value of $t_n = \frac{\bar{x}_n - \mu_0}{s/\sqrt{n}}$ probable regarding the distribution $\mathcal{T}(n-1)$

Illustration

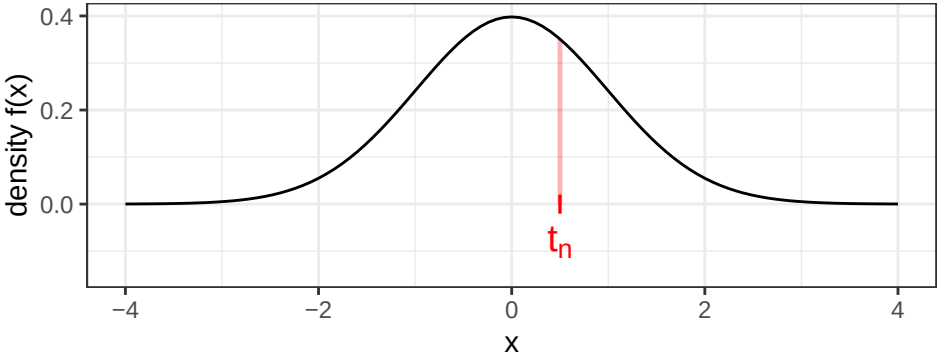
Density of the distribution $\mathcal{T}(n-1)$ with $n = 100$



Improbable value t_n

Illustration

Density of the distribution $\mathcal{T}(n-1)$ with $n = 100$



Probable value t_n

Comments

The testing procedure follows the protocol of **experimental science**

1. Formulate an hypothesis *a priori*
2. Check if the observations confirms or rejects the hypothesis

For any tests, $\mathcal{H}_0$ is favored because assumed to be true to construct the test.

Keeping $\mathcal{H}_0$ does not mean that $\mathcal{H}_0$ is true, a more precise experiment could allow to reject $\mathcal{H}_0$ (example later).

Only rejecting $\mathcal{H}_0$ is informative and significant.

Analogy with the french justice system: every defendant is assumed to be innocent ($\mathcal{H}_0$) until convicted (reject $\mathcal{H}_0$) because proven guilty, or declared innocent (keep $\mathcal{H}_0$) because of lack of proof of guilt. Does the absence of proof of guilt means that the

Error made by a test

- **Type I error rate**[1]: $\alpha = \mathbb{P}(\text{reject } \mathcal{H}_0 \,|\, \mathcal{H}_0 \text{ true})$
- **Confidence level**[2] of the test :$1 - \alpha = \mathbb{P}(\text{keep } \mathcal{H}_0 \,|\, \mathcal{H}_0 \text{ true})$
- **Type II error rate**[3]: $\beta = \mathbb{P}(\text{keep } \mathcal{H}_0 \,|\, \mathcal{H}_1 \text{ true})$
- **Power**[4] of the test: $1 - \beta = \mathbb{P}(\text{reject } \mathcal{H}_0 \,|\, \mathcal{H}_1 \text{ true})$

[1] "*Risque de 1ère espèce*"
[2] "*Seuil de confidence*"
[3] "*Risque de 2nde espèce*"
[4] "*Puissance*"

Summary

		Decision	
		keep $\mathcal{H}_0$	reject $\mathcal{H}_0$
Reality	$\mathcal{H}_0$ true	**Correct decision** Probability $= 1 - \alpha$	**Type I error** Probability $= \alpha$
	$\mathcal{H}_1$ true	**Type II error** Probability $= \beta$	**Correct decision** Probability $= 1 - \beta$

How to control type I and II risk?

- **Type I risk** α is set *a priori* to define the keep/reject $\mathcal{H}_0$ criterion (that we will define later)
  - Type I risk is controlled by the choice of α
  - the α value quantifies the consequence of wrongly rejecting $\mathcal{H}_0$
  - Generally we take $\alpha = 5\%$ (**!! arbitrary choice !!**)

- **Type II risk** β (or power $1 - \beta$) is more difficult to control (because directly related to $\mathcal{H}_1$)
  - Estimation with resampling and randomization
  - Characterization for specific cases of alternate hypothesis $\mathcal{H}_1$

Region of acceptance and region of rejection

We can use the distribution of the statistic T_n under $\mathcal{H}_0$ to find a **region** R such that

$$\mathbb{P}(T_n \in R) = 1 - \alpha$$

This region is called **region of acceptance** and we keep $\mathcal{H}_0$ if the estimate $t_n \in R$ i.e. if the value t_n is probable according to the distribution of T_n.

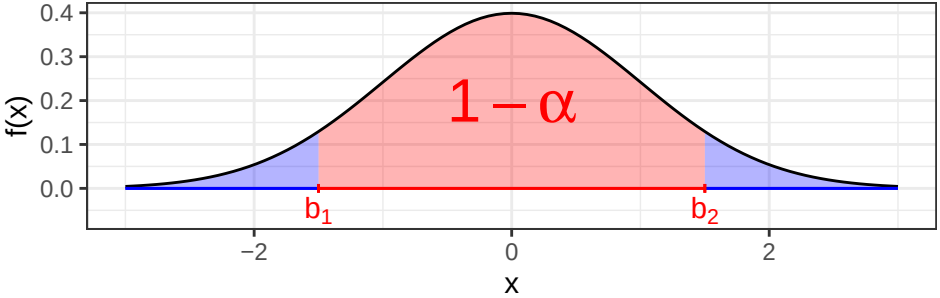
If $t_n \notin R$, we reject $\mathcal{H}_0$ (improbable value under $\mathcal{H}_0$), the region R^c (complementary of R) is called the **region of rejection**.

The acceptance region can be a bilateral or unilateral **interval** (depending on the application).

Bilateral test

The region of acceptance is a **bilateral interval**: $R = [b_1\,;\,b_2]$ where the boundaries b_1 and b_2 are defined such that

$$\mathbb{P}(b_1 \leq T_n \leq b_2) = 1 - \alpha$$



- In red (—): region of acceptance $R = [b_1\,;\,b_2]$
- In blue (—): region of rejection $R^c =]-\infty\,;\,b_1[\,\cup\,]b_2\,;\,+\infty[$

Unilateral test

The region of acceptance is a **unilateral interval**. It can be

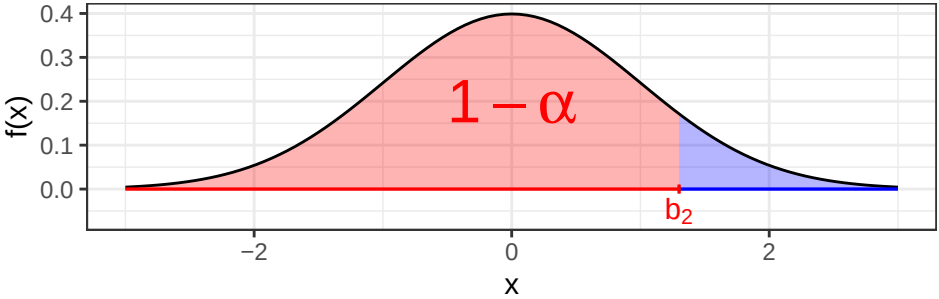
- $R =]-\infty\,;\,b_2]$ where the boundary b_2 is defined such that

$$\mathbb{P}(T_n \leq b_2) = 1 - \alpha$$

- $R = [b_1\,;\,+\infty[)$ where the boundary b_1 is defined such that

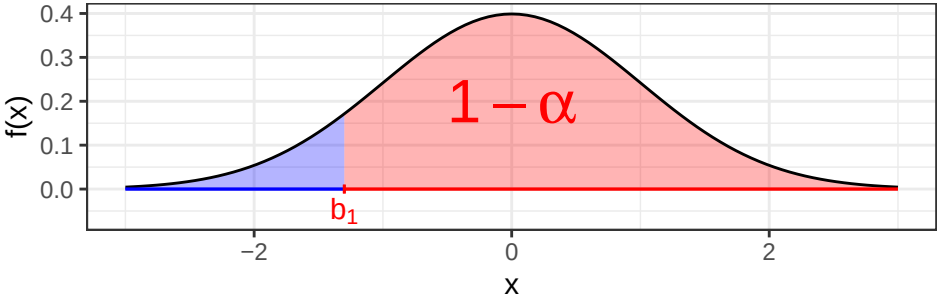
$$\mathbb{P}(b_1 \leq T_n) = 1 - \alpha$$

Unilateral test



- In red (—): region of acceptance $R =]-\infty\,;\,b_2]$
- In blue (—): region of rejection $R^c =]b_2\,;\,+\infty[$

Unilateral test



- In red (—): region of acceptance $R = [b_1\,;\,+\infty[$
- In blue (—): region of rejection $R^c =]-\infty\,;\,b_1[$

Region of rejection / Critical region

- Reminder: the **region of rejection** is the set of values for the test statistic T_n for which the null hypothesis is rejected.

- **Critical value** = the threshold value delimiting the regions of acceptance and rejection for the test statistic T_n.

  In the previous examples, the critical values were b_1 and b_2 for a bilateral interval, b_1 or b_2 for a unilateral interval.

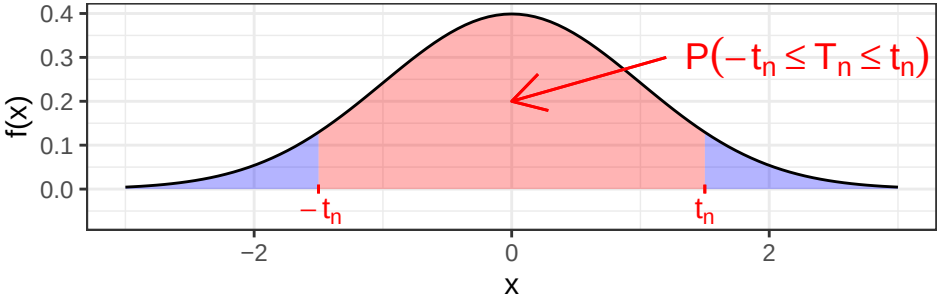
- ***p*-value** = the probability of observing a result at least as extreme as the test statistic value t_n (assuming the null hypothesis is true)

$$p\text{-value} = \mathbb{P}(|T_n| > |t_n|)$$

p-value

For a **bilateral test**

$$p\text{-value} = \mathbb{P}(T_n < -t_n \text{ or } T_n > t_n) = 1 - \mathbb{P}(-t_n \leq T_n \leq t_n)$$

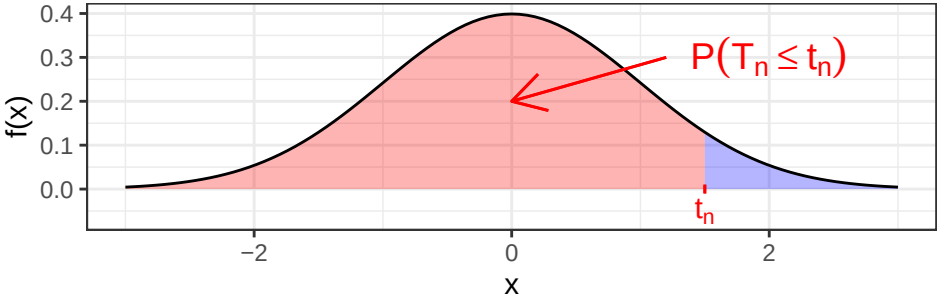


p-value = area in blue

p-value

For a **unilateral test**

$$p\text{-value} = \mathbb{P}(T_n > t_n) = 1 - \mathbb{P}(T_n \leq t_n)$$

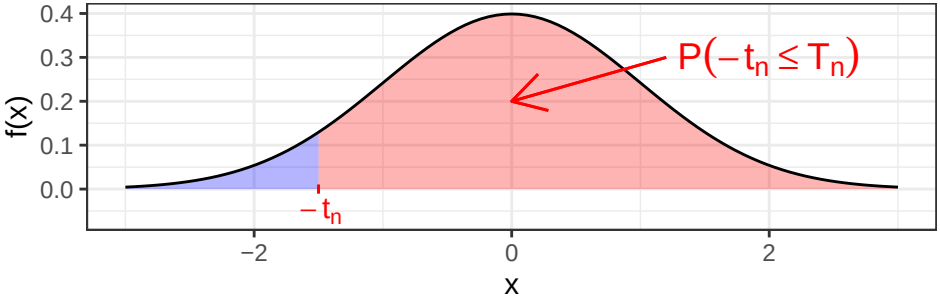


p-value = area in blue

p-value

For a **unilateral test** (other side)

$$p\text{-value} = \mathbb{P}(T_n < -t_n) = 1 - \mathbb{P}(-t_n \leq T_n)$$



p-value = area in blue

Procedure to control type I and type II risks

- **Classical approach:** α is fixed *a priori*.
- **Modern approach:** use a computer
  - α is **not** fixed *a priori*.
  - We rather user the notion the **critical probability** (p-value) to determine the level of significance[1] of the test.
- **Note:** if we **decrease** α (i.e. Type I error)
  1. we reject $\mathcal{H}_0$ only for more extreme values of T_n
  2. consequence 1: we accept less probable results regarding $\mathcal{H}_0$
  3. consequence 2: we **increase** the risk to wrongly keep $\mathcal{H}_0$ (i.e Type II error)

[1] "*niveau de significativité*"

Result of the test based on p-value

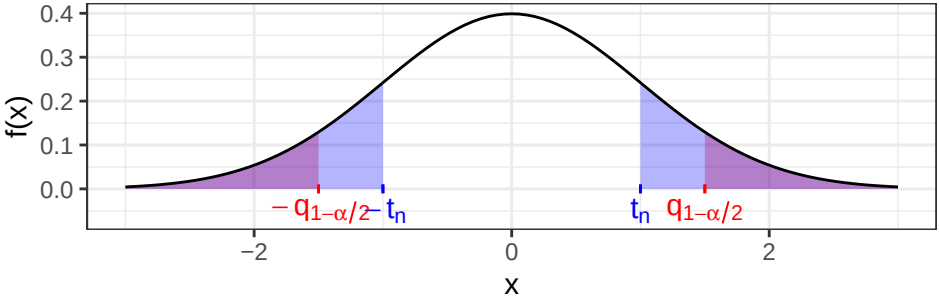
General rule[1]

- If p-value $> \alpha$ then we accept $\mathcal{H}_0$ (**!! not significant !!**)
- If p-value $\leq \alpha$ then we reject $\mathcal{H}_0$ (**!! significant !!**)

[1] !! VERY IMPORTANT !!

Result of the test based on the p-value (bilateral test)

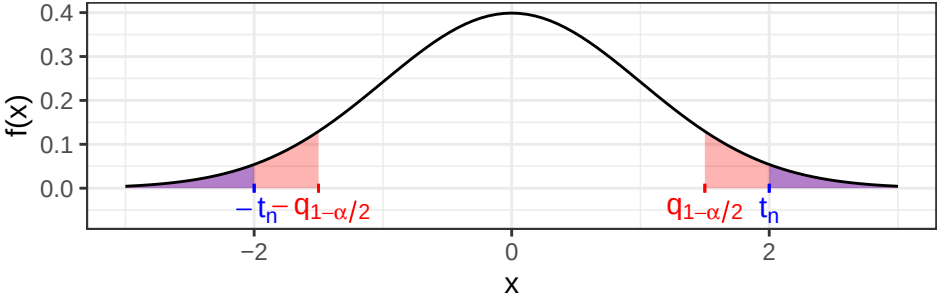
- If p-value $> \alpha$ then we **accept** $\mathcal{H}_0$
- equivalent to $t_n \in$ **region of acceptation**



- Area in blue $=$ p-value $=$ $\mathbb{P}(|T_n| > |t_n|)$
- Area in red $=$ $\alpha = \mathbb{P}(|T_n| > |q_{1-\alpha/2}|)$
- q_α $=$ quantile of the distribution of T_n at level α
- Region of acceptation $R = [-q_{1-\alpha/2}\,;\, q_{1-\alpha/2}]$

Result of the test based on the p-value (bilateral test)

- If p-value $\leq \alpha$ then we **reject** $\mathcal{H}_0$
- equivalent to $t_n \notin$ **region of acceptation**



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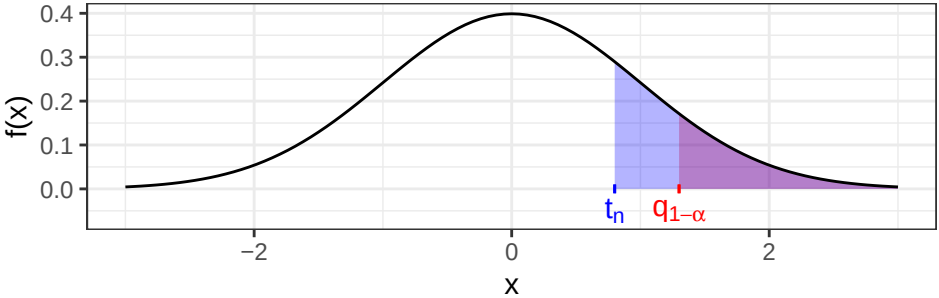
- q_α $=$ quantile of the distribution of T_n at level α
- Region of acceptation $R = [-q_{1-\alpha/2}\,;\, q_{1-\alpha/2}]$

Result of the test based on the p-value (unilateral test)

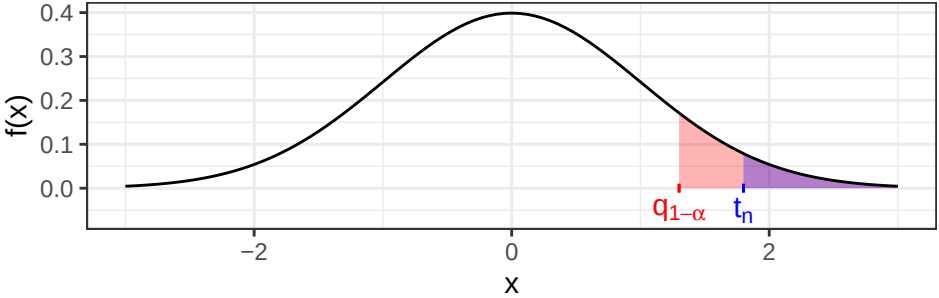
- If p-value $> \alpha$ then we **accept** $\mathcal{H}_0$
- equivalent to $t_n \in$ **region of acceptation**



- Area in blue $=$ p-value $=$ $\mathbb{P}(T_n > t_n)$
- Area in red $= \alpha = \mathbb{P}(T_n > q_{1-\alpha})$
- $q_\alpha =$ quantile of the distribution of T_n at level α
- Region of acceptation $R = [-\infty\,;\, q_{1-\alpha}]$

Result of the test based on the p-value (unilateral test)

- If p-value $\leq \alpha$ then we **reject** $\mathcal{H}_0$
- equivalent to $t_n \notin$ **region of acceptation**



- Area in blue $=$ p-value $=$ $\mathbb{P}(T_n > t_n)$
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- $q_\alpha =$ quantile of the distribution of T_n at level α
- Region of acceptation $R = [-\infty\,;\, q_{1-\alpha}]$

Hypothesis testing in practice

- The **test statistic** T_n should have an **explicit** probability distribution when **assuming $\mathcal{H}_0$ true**. It can be
  - a difference between headcounts expected under $\mathcal{H}_0$ and empirical headcounts[1] (e.g. χ^2 test of independence)
  - the estimator of a parameter with a probability distribution known under $\mathcal{H}_0$ (e.g. T_n can be the empirical mean)
  - the number of expected successes under $\mathcal{H}_0$ (e.g. T_n can be the empirical frequency)
- **A test may concern:**
  - a single population (e.g. proportion of smoking person $> 20\%$ in France?)
  - two populations (e.g. survival under treatment with drug 1 is better than survival under treatment with drug 2)
  - or more than 2 populations

[1] "*écart entre effectifs attendus sous $\mathcal{H}_0$ et effectifs observés*"

Outline

Introduction to tests

Tests on a single population

Tests on two populations

Parametric tests on a single population

- **Test regarding the value of a parameter of a probability distribution**
- Example: mean, variance, proportion

Parametric tests regarding the theoretical mean/expectation

Let $(X_1, \ldots, X_n)$ be a n-sample associated to a random variable X of unknown mean μ.

- We fix a value μ_0.
- We want to test $\mathcal{H}_0$: "$\mu = \mu_0$" **versus** $\mathcal{H}_1$: "$\mu \neq \mu_0$"
- We will reject $\mathcal{H}_0$ if the difference $\mu - \mu_0$ is too large (bilateral test)
  $\rightarrow$ characterize the sample variation $\mu - \mu_0$ under $\mathcal{H}_0$
- μ is **unknown**, we use the empirical mean estimator $\widehat{\mu} = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ as **test statistic**

Parametric tests regarding the theoretical mean/expectation (when σ^2 is known)

When the variance σ^2 is **known**, we have $\widehat{\mu} \sim \mathcal{N}(\mu, \sigma^2/n)$ (because X Gaussian or large n)

If $\mathcal{H}_0$ is assumed to be true (i.e. $\mu = \mu_0$), we can use the following statistic $\widehat{\mu} \sim \mathcal{N}(\mu_0, \sigma^2/n)$

We have then

$$\mathbb{P}\left(-z_{1-\alpha/2} \leq \frac{\widehat{\mu} - \mu_0}{\sigma/\sqrt{n}} \leq z_{1-\alpha/2} \middle| \mathcal{H}_0\right) = 1 - \alpha$$

where $z_{1-\alpha/2}$ is the quantile of level $1 - \alpha/2$ for the standard Gaussian distribution.

Parametric tests regarding the theoretical mean/expectation (when σ^2 is known)

Test

- at the confidence level $1-\alpha$, we **keep** $\mathcal{H}_0$ if

$$\frac{\widehat{\mu}-\mu_0}{\sigma/n} \in \left[-z_{1-\alpha/2}\,;\, z_{1-\alpha/2}\right]$$

- or equivalently, at risk α, we **reject** $\mathcal{H}_0$ if

$$\frac{\widehat{\mu}-\mu_0}{\sigma/n} < -z_{1-\alpha/2} \quad \text{or} \quad \frac{\widehat{\mu}-\mu_0}{\sigma/n} > z_{1-\alpha/2}$$

Parametric tests regarding the theoretical mean/expectation (when σ^2 is unknown)

When the variance σ^2 is **unknown**, we have $\widehat{\mu} \sim \mathcal{N}(\mu, \sigma^2/n)$

We can estimate σ^2 by $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$ and we have (reminder)
$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1)$

Consequence:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$$

i.e. Student's distribution with $n-1$ degrees of freedom

Parametric tests regarding the theoretical mean/expectation (when σ^2 is unknown)

If $\mathcal{H}_0$ is assumed to be true (i.e. $\mu = \mu_0$), we can use the following statistic $\frac{\widehat{\mu}-\mu_0}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$

We have then

$$\mathbb{P}\left(-t_{1-\alpha/2;n-1} \leq \frac{\widehat{\mu}-\mu_0}{S/\sqrt{n}} \leq t_{1-\alpha/2;n-1} \middle| \mathcal{H}_0\right) = 1-\alpha$$

where $t_{1-\alpha/2;n-1}$ is the quantile of level $1-\alpha/2$ for the Student's $\mathcal{T}(n-1)$ distribution (reminder $t_{1-\alpha/2;n-1} \approx z_{1-\alpha/2}$ for large n).

Parametric tests regarding the theoretical mean/expectation (when σ^2 is unknown)

Test

- at the confidence level $1-\alpha$, we **keep** $\mathcal{H}_0$ if

$$\frac{\widehat{\mu}-\mu_0}{s/n} \in \left[-t_{1-\alpha/2;n-1}\,;\; t_{1-\alpha/2;n-1}\right]$$

- or equivalently, at risk α, we **reject** $\mathcal{H}_0$ if

$$\frac{\widehat{\mu}-\mu_0}{s/n} < -t_{1-\alpha/2;n-1} \quad \text{or} \quad \frac{\widehat{\mu}-\mu_0}{s/n} > t_{1-\alpha/2;n-1}$$

Problematic cases

If n is small (and X not Gaussian), two possibilities:

- Derive the exact distribution of $\widehat{\mu}$ under $\mathcal{H}_0$
- Use non-parametric tests

Review of some parametric tests

Bilateral $\mathcal{T}$-test for the mean/expectation (c.f. previous example)

- **Pre-requisite:** X of mean μ and unknown variance is Gaussian (hence $\bar{X}_n$ is Gaussian) or $\bar{X}_n$ is asymptotically Gaussian (central limit theorem)

- **Hypothesis:** "$\mathcal{H}_0$: $\mu = \mu_0$" versus "$\mathcal{H}_1$: $\mu \neq \mu_0$"

- **Test statistic:** $T_n = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$ under $\mathcal{H}_0$

- **Region of acceptation:** $t_n \in [-t_{1-\alpha/2;n-1}\,;\, t_{1-\alpha/2;n-1}]$?
  with $t_{1-\alpha/2;n-1}$ quantile of level $1-\alpha/2$ of the distribution $\mathcal{T}(n-1)$

- p-value: $P(|T_n| > |t_n|) = 1 - \mathbb{P}(-t_n \leq T_n \leq t_n)$

Unilateral $\mathcal{T}$-test for the mean/expectation

- **Pre-requisite:** X of mean μ and unknown variance σ^2 is Gaussian (hence $\bar{X}_n$ is Gaussian) or $\bar{X}_n$ is asymptotically Gaussian (central limit theorem)

- **Hypothesis:** "$\mathcal{H}_0$: $\mu = \mu_0$" versus "$\mathcal{H}_1$: $\mu > \mu_0$" ou $\mathcal{H}_1$: "$\mu < \mu_0$"

- **Test statistic:** $T_n = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$ under $\mathcal{H}_0$

- **Region of acceptation:** $t_n \in [-\infty\,;\, t_{1-\alpha;n-1}]$? ou $t_n \in [-t_{1-\alpha;n-1}\,;\, +\infty]$? with $t_{1-\alpha;n-1}$ quantile of level $1-\alpha$ of the distribution $\mathcal{T}(n-1)$

- p-value: $P(T_n > t_n) = 1 - \mathbb{P}(T_n \leq t_n)$ ou $P(T_n < -t_n)$

Bilateral test for a proportion

- **Pre-requisite:** $X \sim \mathcal{B}(p)$, $n > 100$, $np_0 > 5$, $n(1-p_0) > 5$ i.e. $5\% < p_0 < 95\%$ (to use central limit theorem)

- **Hypothesis:** "$\mathcal{H}_0$: $p = p_0$" versus "$\mathcal{H}_1$: $p \neq p_0$"

- **Test statistic:** $Z_n = \frac{\widehat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim \mathcal{N}(0,1)$ under $\mathcal{H}_0$

- **Region of acceptation:** $z_n \in [-z_{1-\alpha/2}\,;\, z_{1-\alpha/2}]$? with z_n observed value for Z_n and $z_{1-\alpha/2}$ quantile of level $1-\alpha/2$ of the distribution $\mathcal{N}(0,1)$

- p-value: $P(|Z_n| > |z_n|) = 1 - \mathbb{P}(-z_n \leq Z_n \leq z_n)$

Bilateral test for a proportion

- if n is small? we know the exact distribution: under $\mathcal{H}_0$

$$n\,\widehat{p} \sim \mathcal{B}(n, p_0)$$

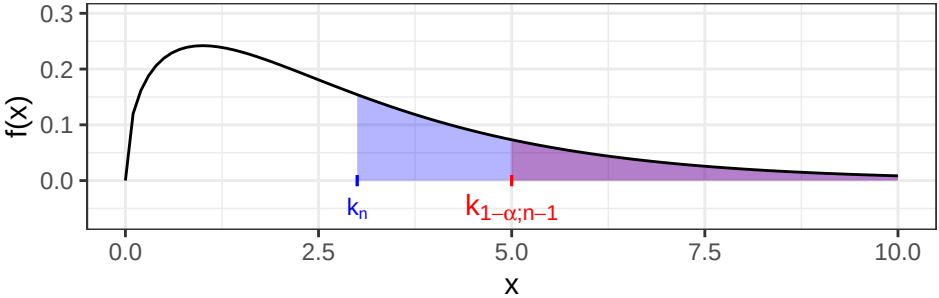
Unilateral[1] test for the variance

- **Pre-requisite:** **X is Gaussian** of mean μ and unknown variance σ^2
- **Hypothesis:** **"$\mathcal{H}_0$: $\sigma = \sigma_0$"** versus "$\mathcal{H}_1$: $\sigma > \sigma_0$" (large variance)
- **Test statistic:** $K_n = (n-1)\frac{S^2}{{\sigma_0}^2} \sim \chi^2(n-1)$ under $\mathcal{H}_0$ with $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$
- **Region of acceptation:** $k_n \in [0\,;\, k_{1-\alpha;n-1}]$? with k_n observed value for K_n and $k_{1-\alpha;n-1}$ quantile of level $1-\alpha$ of the distribution $\chi^2(n-1)$
- p-value: $P(K_n > k_n) = 1 - \mathbb{P}(K_n \leq k_n)$

[1] Useless to check for a lower bound on the variance, the lower the variance, the better

Unilateral[1] test for the variance

- If p-value $> \alpha$ then we **accept** $\mathcal{H}_0$
- equivalent to $k_n \in$ **region of acceptation**

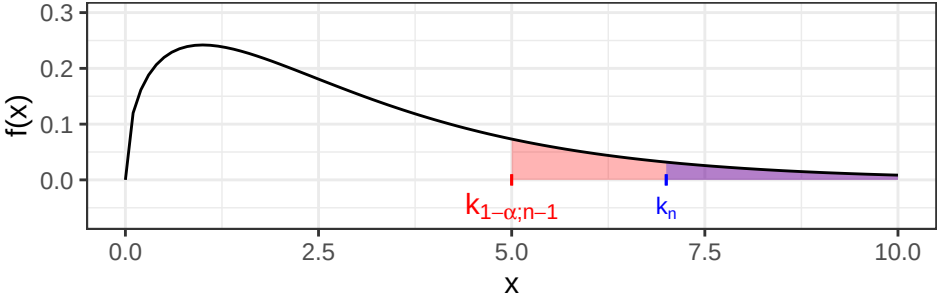


- Area in blue $=$ p-value $=$ $\mathbb{P}(K_n > k_n)$
- Area in red $=$ $\alpha = \mathbb{P}(K_n > k_{1-\alpha;n-1})$

- $k_{\alpha;n-1}$ $=$ quantile of $\chi^2(n-1)$ at level α
- Region of acceptation $R = [0;\ k_{1-\alpha;n-1}]$

Unilateral[1] test for the variance

- If p-value $\leq \alpha$ then we **reject** $\mathcal{H}_0$
- equivalent to $k_n \notin$ **region of acceptation**



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- $k_{\alpha;n-1}$ $=$ quantile of $\chi^2(n-1)$ at level α
- Region of acceptation $R = [0;\ k_{1-\alpha;n-1}]$

Outline

Introduction to tests

Tests on a single population

Tests on two populations

Parametric tests on two populations

- Example: comparison of mean or comparison of variance

- For more details, see `https://www.math.univ-toulouse.fr/~besse/Wikistat/pdf/st-l-inf-tests.pdf` (section 4, in french)

Non-parametric test

- Independence test (χ^2)
- Adjustment test (χ^2)
- Normality test: e.g. Shapiro-Wilk test (c.f. tutorial)

Example: independence test

- **Question:** is smoking dependent on the genre in youth population ?

- **Hypothesis:**

  "$\mathcal{H}_0$: the *genre* and *smoking* variables are independent"

  versus

  "$\mathcal{H}_1$: *genre* and *smoking* are not independent"

We do a study, i.e. interview $n = 360$ young persons, and get the following results (headcounts):

		Smoking		Total
		Yes	No	
Genre	Female	50	160	210
	Male	70	80	150
Total		120	240	360

Independence test: empirical frequencies

We note F for female and M for male, Sm for smoking and nSm for not smoking.

Based on the headcounts, we can define the **empirical frequencies** in the sample:

- Female smoker:
  $\widehat{P}(F \text{ and } Sm) = \frac{50}{360}$
- Female non-smoker:
  $\widehat{P}(F \text{ and } nSm) = \frac{160}{360}$
- Male smoker:
  $\widehat{P}(M \text{ and } Sm) = \frac{70}{360}$
- Male non-smoker:
  $\widehat{P}(M \text{ and } nSm) = \frac{80}{360}$

- Female: $\widehat{P}(F) = \frac{210}{360}$
- Male: $\widehat{P}(M) = \frac{150}{360}$
- Smoker: $\widehat{P}(Sm) = \frac{120}{360}$
- Non-Smoker: $\widehat{P}(nSm) = \frac{240}{360}$

Independence test: headcounts

- Notations:
  - condition "*Genre*" indexed by i (i.e. $i \in \{F, M\}$)
  - condition "*Smoking status*" indexed by j (i.e. $j \in \{Sm, nSm\}$)
- Total sample size $= n$
- **Headcounts** for combination of conditions *Genre* i and *Smoking status* $j = n_{ij}$
- Total headcounts for condition *Genre* i: $n_{i+} = \sum_j n_{ij}$
- Total headcounts for condition *Smoking status* j: $n_{+j} = \sum_i n_{ij}$

Independence test: headcounts and empirical frequencies

Bases on headcounts, **empirical frequencies** are:

- for combination of conditions *Genre i* and *Smoking status j*: $\frac{n_{ij}}{n}$
- for condition *Genre i*: $\frac{n_{i+}}{n}$
- for condition *Smoking status j*: $\frac{n_{+j}}{n}$

Independence test: independence?

If independence between *Genre* and *Smoking status*, then regarding **unknown theoretical probabilities** we would have:

- $\mathbb{P}(F \text{ and } Sm) = \mathbb{P}(F) \times \mathbb{P}(Sm)$
- $\mathbb{P}(H \text{ and } Sm) = \mathbb{P}(H) \times \mathbb{P}(Sm)$
- $\mathbb{P}(F \text{ and } nSm) = \mathbb{P}(F) \times \mathbb{P}(nSm)$
- $\mathbb{P}(H \text{ and } nSm) = \mathbb{P}(H) \times \mathbb{P}(nSm)$

Consequence: for **empirical frequencies** we would expect that:

- $\widehat{\mathbb{P}}(F \text{ and } Sm) \approx \widehat{\mathbb{P}}(F) \times \widehat{\mathbb{P}}(Sm)$
- $\widehat{\mathbb{P}}(H \text{ and } Sm) \approx \widehat{\mathbb{P}}(H) \times \widehat{\mathbb{P}}(Sm)$
- $\widehat{\mathbb{P}}(F \text{ and } nSm) \approx \widehat{\mathbb{P}}(F) \times \widehat{\mathbb{P}}(nSm)$
- $\widehat{\mathbb{P}}(H \text{ and } nSm) \approx \widehat{\mathbb{P}}(H) \times \widehat{\mathbb{P}}(nSm)$

i.e. $\frac{n_{ij}}{n} = \frac{n_{i+}}{n} \times \frac{n_{+j}}{n}$ i.e. $n_{ij} = \frac{n_{i+} n_{+j}}{n}$

Independence test: theoretical headcounts

Using the independence-based relation, we can compute **theoretical headcounts** n^*_{ij} under the hypothesis of independence between Genre and Smoking variable, i.e.

$$n^*_{ij} = \frac{n_{i+} n_{+j}}{n}$$

We get the following table:

		Smoking		Total
		Yes	No	
Genre	Female	**70**	**140**	210
	Male	**50**	**100**	150
Total		120	240	360

Independence test: χ^2 test statistic

- **Test statistics:**

$$K2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{\left(n_{ij} - \frac{n_{i+}\, n_{+j}}{n}\right)^2}{\frac{n_{i+}\, n_{+j}}{n}}$$

with I and J the respective numbers of conditions for the two variables (here *Genre* and *Smoking status* each have 2 conditions, thus $I = J = 2$).

- Under $\mathcal{H}_0$: $K2 \sim \chi^2((I-1) \times (J-1))$

- **Note:**

$$K2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(\text{"Empirical headcounts"} - \text{"Theoretical headcounts"})^2}{\text{"Theoretical headcounts"}}$$

Independence χ^2 test

- **Unilateral test:** if the observation $k2$ of the statistic $K2$ is too large then the deviation between empirical and theoretical headcounts is too large and we **reject** $\mathcal{H}_0$ (**meaning that the two variables are significantly non independent**).

- **Acceptation region:** $[0\,;\, k_{1-\alpha;(I-1)(J-1)}]$ with $k_{1-\alpha;(I-1)(J-1)}$ the quantile of the distribution $\chi^2((I-1)\times(J-1))$ at level $1-\alpha$.

- p-value $= \mathbb{P}(K2 > k2) = 1 - \mathbb{P}(K2 < k2)$

Example: result of the test? (exercise)

- Degrees of freedom:
- $k2 = $
- Quantile:
- p-value =

Example: result of the test

Questions?